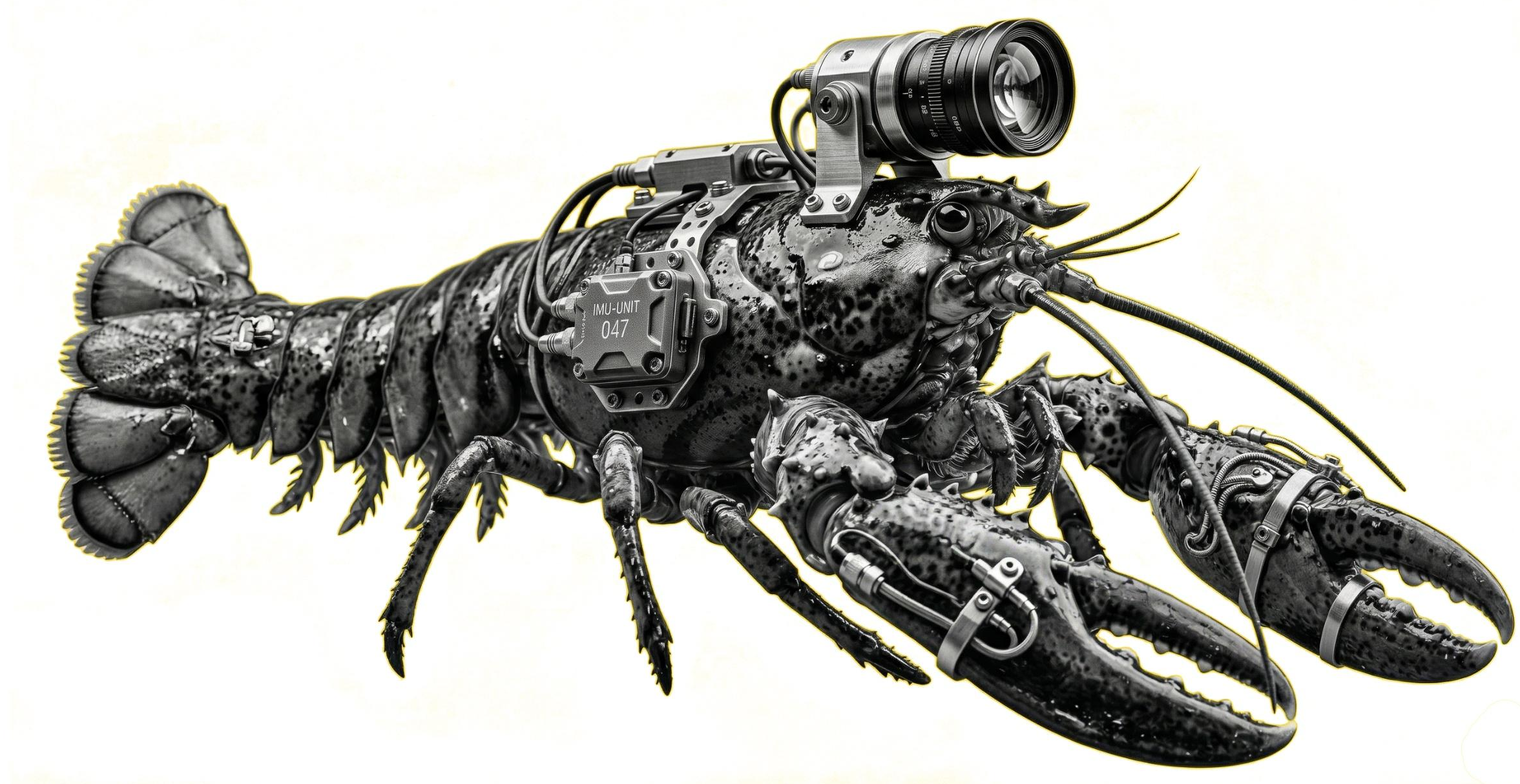


Physical-World Observation for Personal AI Agent Personalization

Combining Egocentric Sensing with Transparent Agent Memory.

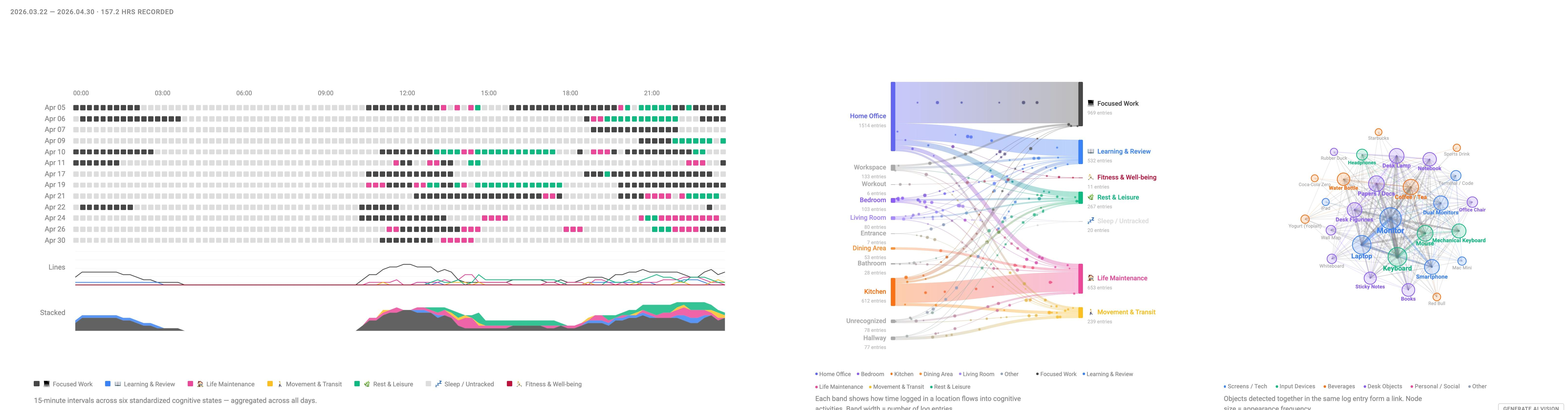
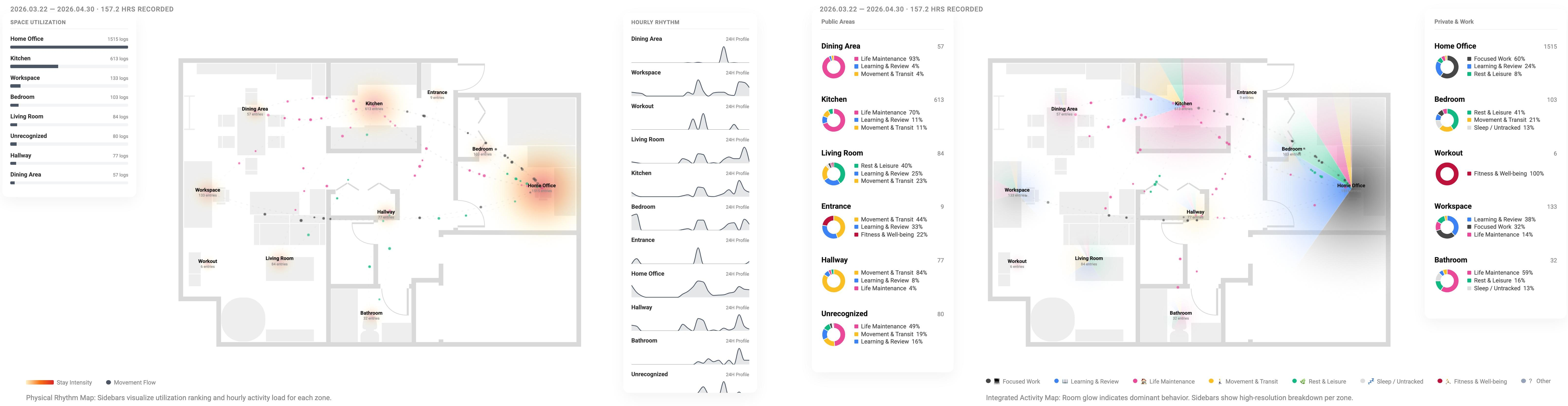
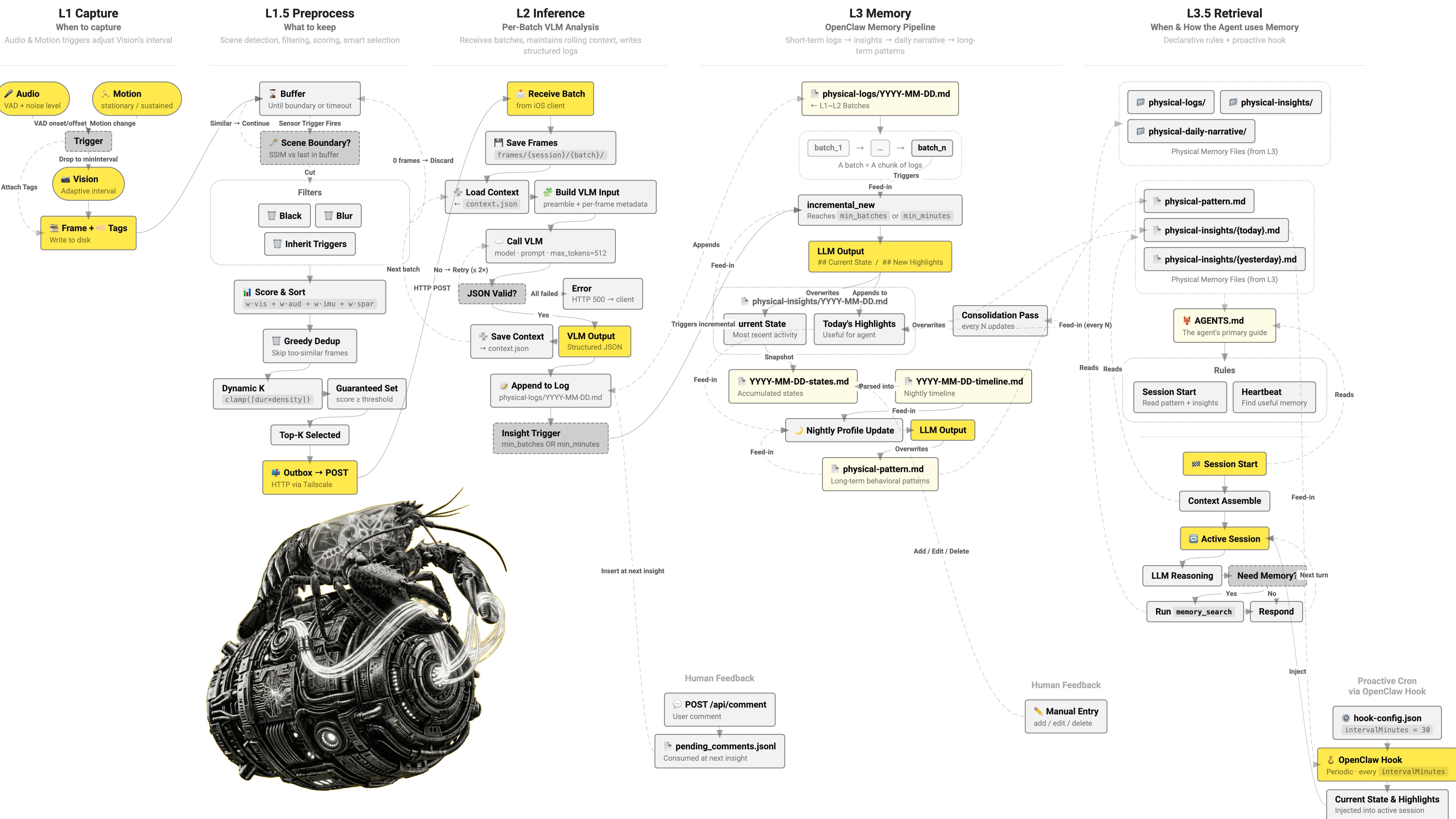
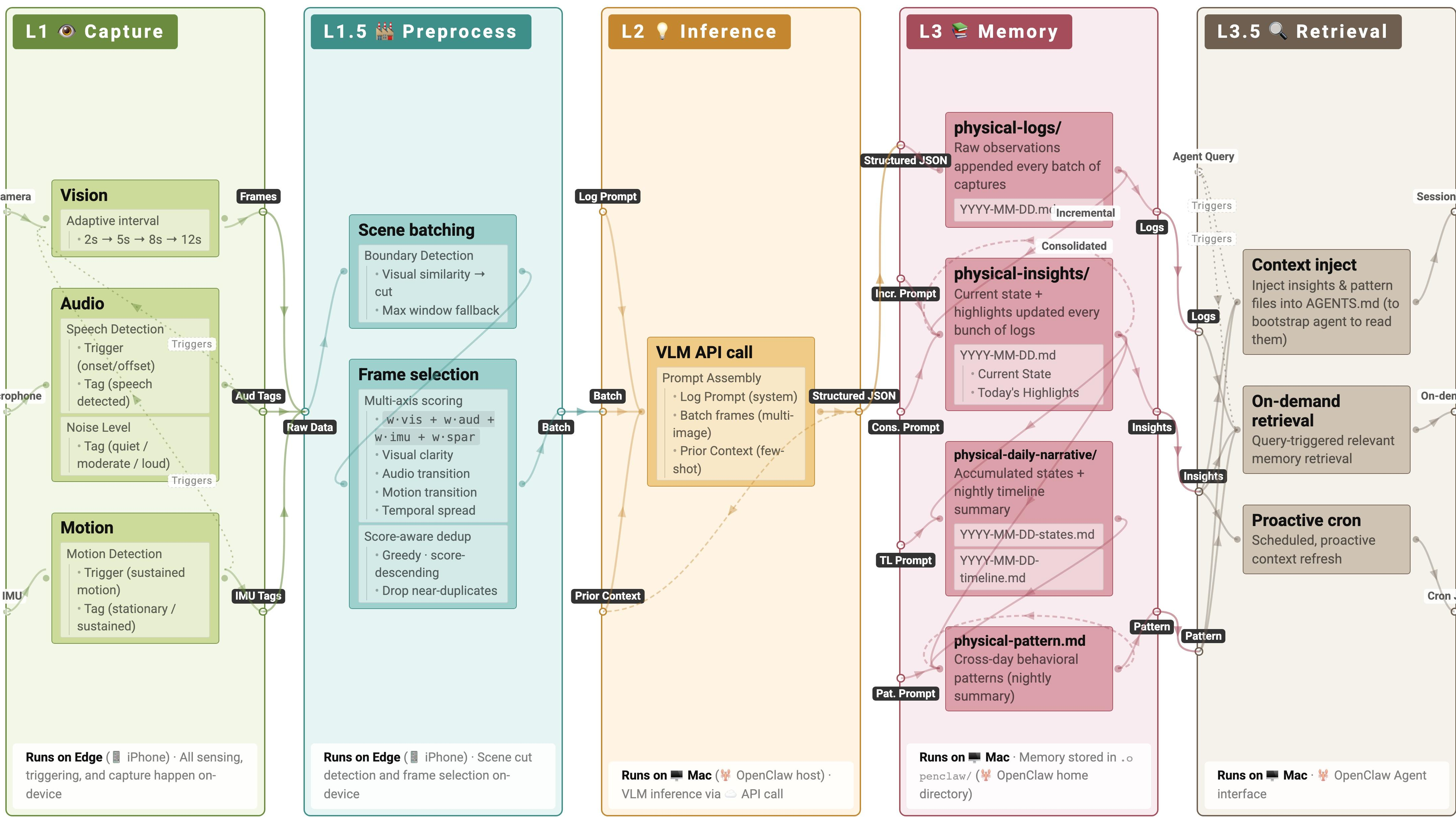
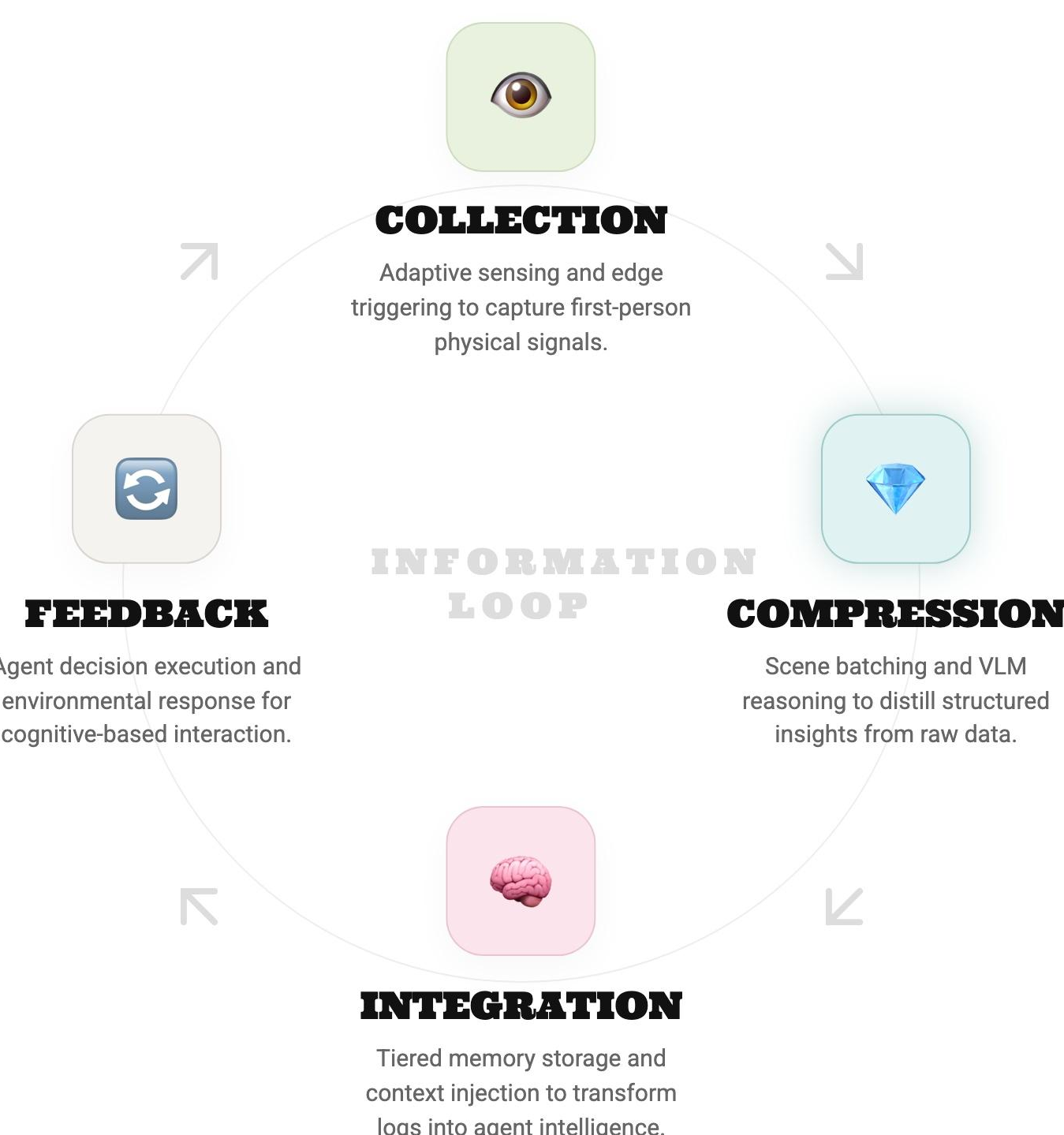


Research Questions

- RQ1** What **physical-world information** becomes visible through egocentric perception that remains invisible through digital traces alone?
- RQ2** In what **task & interaction contexts** does this physical-world information meaningfully improve personalization, and when does it introduce noise or redundancy?
- RQ3** What **design requirements** emerge for integrating continuous egocentric sensing into existing agent memory architectures?

Contributions

- C1** A two-week **autoethnographic deployment study** examining what physical-world information egocentric perception makes visible and when it improves agent personalization.
- C2** A **system artifact** integrating egocentric vision into OpenClaw's markdown-based memory architecture.
- C3** **Design implications** for physical-world memory integration, covering storage granularity, retrieval visibility, and privacy trade-offs.



AGENT INSIGHT

"Beverage rotation synced to workload: Espresso (design/app pivot) → soda/orange cream (deep focus) → tea (log review) → latte (image browsing)."

SENSORY PATTERN

Observed in 40 of 44 late sessions

AGENT INSIGHT

"Late-night kitchen looping consolidated: Desk ↔ kitchen ↔ snacking ↔ beverage ↔ return."

SPATIAL LOGISTICS

Observed in 41 of 44 late cycles

AGENT INSIGHT

"Paper-digital bridge sustained: Printed maps clipped to desk, sticky-note archiving, OCR/scan placement, colored sticky pads, hospital meal tray protocols."

WORKFLOW BRIDGING

Observed in 49 of 53 recent workdays

AGENT INSIGHT

"Persistent delayed sleep-wake cycle: Rest ~01:00-01:30, wake ~11:20."

RHYTHM DYNAMICS

Observed 57 of 58 days

AGENT INSIGHT

"Desk periphery expansion: Dual-laptop desk ↔ reclined bedroom → kitchen → hybrid studio integration."

WORKSPACE EXPANSION

Observed 22 of 23 late nights

AGENT INSIGHT

"Analog-digital harmonization daily: Green notebook present during snacking; sticky-note task markers on refrigerator; dual-laptop editing of Physical-World Observation document."

ANALOG-DIGITAL HARMONIZATION

Observed in 46 of 46 recent evenings